Impact of Polarization Basis on Wind and Wave Parameters Estimation Using the Azimuth Cutoff From GF-3 SAR Imagery

Liwei Bao, Xi Zhang, Chenghui Cao, Xiaochen Wang, Yongjun Jia, Gui Gao, *Member, IEEE*, Yi Zhang, Yong Wan, *Member, IEEE*, and Jie Zhang

***Abstract*— The azimuth cutoff wavelength of synthetic aperture radar (SAR) is an important parameter for retrieval of sea surface wind and wave. Earlier studies have fully demonstrated the substantial dependence of azimuth cutoff wavelength on polarization, but the present studies only focus on horizontal and vertical (H-V) linear polarization bases [horizontal-horizontal (HH), horizontal-vertical (HV)/vertical-horizontal (VH), and vertical-vertical (VV)] without considering the effects of other polarization bases (e.g., linear rotated, circular, and elliptical polarization). Benefiting from the quad-polarization advantage of GaoFen-3 SAR wave mode data and the support of polarization basis transformation theory, this study used 4648 SAR data to study the correlation between cutoff wavelength and wind and wave parameters (e.g., significant wave height and wind speed) under different polarization bases and analyzed the variation of correlation coefficient caused by polarization basis change. Finally, the results were applied to evaluate the performance of wind and wave parameters retrieval. The results of the study show that the azimuth cutoff is strongly dependent on the polarization state of the electromagnetic wave. The azimuth cutoff wavelength under the elliptical polarization bases has a higher correlation with wind and wave than that under H-V linear, circular, and linear rotated polarization bases. Using the azimuth cutoff wavelength of the elliptical polarization bases can significantly improve the retrieval accuracy of wind and wave parameters. This study shall enhance the capabilities of polarized SAR systems to precisely derive more ocean surface properties. The result implies that polarization basis is an important factor that must be considered in future ocean SAR studies.**

***Index Terms*— Azimuth cutoff, parameters of wind and wave, polarization basis transformation, synthetic aperture radar (SAR).**

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I. Introduction

WIND and wave parameters are the most basic environmental parameters in the process of ocean dynamics, which is valuable for studying the coupling effect of the ocean and atmosphere [1], analyzing global climate change, and predicting global wind speed (WS) and wave height trends [2]. Synthetic aperture radar (SAR) has the advantages of high resolution, all-weather, and all-time observation and has always been the main method of obtaining wind and wave information. Because SAR adopts a synthetic aperture imaging method, when the ocean waves on the sea surface propagate along the SAR azimuth direction, the effect of the orbital velocity of long waves on the backscattering unit causes additional doppler frequency shift, which leads to an azimuth offset of the imaging surface element on the SAR image, causing the dispersion and aggregation of the scattering surface. When the azimuth offset is greater than the wavelength of a long wave, the short-wave component propagating along the azimuth direction will be blurred. This effect is called "velocity bunching" [3] and will eventually make waves with wavelengths that are less than a certain threshold unobservable by the SAR [4]. This problem in SAR wave observations is generally called the azimuth cutoff phenomenon [5].

In SAR ocean imaging, the cutoff wavelength is an indispensable parameter. A large number of studies have shown that there is a high correlation between the cutoff wavelength, and wind and wave information. Considering the influence of the average incident angle and wind direction angle on the retrieval results, the cutoff wavelength has an approximate linear relationship with the square root of the significant wave height (SWH). Under fully developed sea state conditions, a linear relationship between the cutoff wavelength and WS at a height of 10 m has also been established [6]. Therefore, empirical methods can be used to retrieve information, such as wave orbit velocity [7], WS [8], [9], and SWH [10], [11].

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In addition, the cutoff wavelength has also been used to establish a deep learning model for the SAR observation of ocean waves. Many researchers regard the cutoff wavelength as an important factor in model establishment [12].

Although the cutoff wavelength plays an important role in the observation of wind and wave, it is affected by many factors. First, the cutoff wavelength primarily depends on the velocity bunching modulation. When the wave propagates along the azimuth direction, the displacement generated by the SAR imaging is a product of the range-to-velocity ratio and the radial velocity component of the directional radar. This aspect strictly depends on the velocity distribution of the scatterer in the SAR resolution unit, and it only affects the azimuth direction of the wave. Second, the limited life is an essential attribute of the sea surface scattering element, which is called the intrinsic scene coherence time [5]. When the scene coherence time is less than the SAR acquisition time, this will also lead to azimuth cutoff. In addition, some studies have proved that the cutoff wavelength is not only dependent on the above radar configuration and wave spectrum [3], [13], [14], [15] but also affected by the radar polarization. Li *et al.* [5] found that the cutoff wavelength of a copolarization [horizontal-horizontal (HH) or vertical-vertical (VV)] image is much smaller than cross-polarization [vertical-horizontal (VH)/horizontal-vertical (HV)] image by analyzing RADARSAT-2 quad-polarization SAR data. This is mainly because the scatterers detected by the cross polarization are more sensitive to the influence of variable random motions. In the cross polarization, the life of detected scatterers is shorter, which reduces the coherent integration time between the SAR and ocean waves, resulting in a large cutoff wavelength [16], [17], [18], [19]. Compared with VV polarization, HH polarization is more susceptible to nonpolarized scattering from the wave breaking effect [20], which makes HH polarization more sensitive to random motion than VV polarization [5], and hence, the cutoff wavelength of HH polarization is also relatively large. Moreover, the ability of SAR to obtain wind and wave information is also affected by the polarization channel. The cutoff wavelengths of different polarizations in RADARSAT-2 quad-polarization data have different relationships with the SWH. Ren *et al.* [10] proposed two models (lookup table and empirical) to retrieve SWH using cutoff wavelength in four different polarization channels (HH, HV, VH, and VV). They found that the retrieval accuracy of copolarization is better than cross polarization.

Polarization is important information in microwave remote sensing, the exploration and utilization of polarization diversity have the advantage to better interpret the scattering mechanisms [21], [22]. Electromagnetic waves exist in various polarization states. Horizontal and vertical (H-V) polarization transmission and H-V polarization reception are the most common linear polarization states. In addition to H-V Cartesian linear polarization, the electromagnetic wave has many other polarization forms, such as 45° linear rotated polarization, circular polarization, and elliptical polarization. For example, in RADARSAT Constellation Mission (RCM) compact polarization SAR system, circular polarization emission and linear polarization reception are used to obtain the sea surface information. Zhu *et al.* [23] combined the backscattering information of 45° linear rotated polarization and VV polarization to estimate the wave slope spectrum. In addition, several studies have shown that some polarization states are as effective as linear polarization in detecting wind and waves on the sea surface. Collins *et al.* [24] introduced compact polarization into the SAR deep learning retrieval model and found that the right circular–left circular (RL), right circular–vertical (RV), and right circular–horizontal (RH) polarization states are similar to linear polarization in terms of retrieval performance. In some cases, the retrieval accuracy of RL is better than that of linear polarization. In addition, Zhang *et al.* [25] developed the C-band compact polarization model functions (e.g., CMODRH, CMODRV, and CMODRL, where CMOD is the C-band model geophysical model functions; RH, RV, and so on are polarization modes) to retrieve WS according to the different combinations of transceiver polarization in compact polarization (e.g., RH, RV, and RL). By comparison of the wind retrieval accuracy of a new function with the CMOD under H-V linear polarization (e.g., CMODH5.N), the results showed that the compact polarization can replace the linear polarization in WS retrieval. Moreover, the retrieval results of CMODRR under high WS conditions are substantially better than those of other polarization states.

From the above discussion, it can be inferred that polarization not only has a significant impact on the cutoff wavelength but also affects the retrieval of wind and wave information. It is natural to ask how the polarization state under different polarization bases affects the parameters of wind and wave retrieval, as well as whether there is a regularity. At present, the H-V linear polarization state is widely used in the study of wind and wave information retrieval. However, it is not known if it performs best on all polarization bases. Also, we are interested in knowing whether introducing another polarization base will significantly improve retrieval accuracy. Polarization basis transformation is expected to solve these problems. According to the method of polarization basis transformation [26], the polarization response under any other basis can be obtained by changing the geometric parameters of the polarization basis, which is equivalent to obtaining the scattering information of the target under any polarization bases. Given richer polarization information, we can deepen our understanding of the influence of polarization on the SAR scattering mechanism and, hence, introduce a more appropriate polarization basis for the retrieval of wind and wave parameters.

In this study, 4648 scenes of GF-3 SAR wave mode (WM) data, with a series of preprocessing such as polarization basis transformation, were adopted. Then, the correlation between the azimuth cutoff wavelength, which was retrieved from the WM data under different polarization bases and different wind and wave parameters, was evaluated. After that, the results were used to analyze the influence of different polarization bases on the retrieval of wind and wave parameters. Finally, a comparison of the retrieval accuracy of wind and wave parameters under different polarization bases was used to demonstrate that the appropriate choice of polarization basis will enhance the capabilities of polarized SAR systems to precisely derive wind and wave information.

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TABLE I
INTRODUCTION OF DATA PARAMETERS OF GF-3 SAR WAVE MODE

Parameters	Value
Spatial resolution (Azi^1/Ra^2)	10m/8-12m
Imaging width	5 km
Incident angle	20°-50°
Polarization mode	Quad-polarization (HH, HV, VH, VV)

[1] Azi means Azimuth resolution
[2] Ra means Range resolution

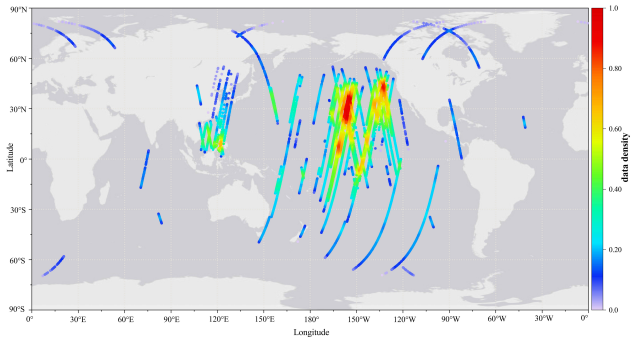

Fig. 1. Spatial distribution diagram of GF-3 SAR WM data.

The structure of this article is given as follows. Section II briefly introduces the dataset and the theoretical foundation. Section III analyzes the correlation between cutoff wavelength, and wind and wave parameters under different polarization bases. Section IV explains the results of Section III with the theoretical model. Section V compares the retrieval accuracy of wind and wave parameters under a partial polarization base based on the findings of Section III. For conclusions and discussions, see Section VI.

II. DATASETS AND METHODS

This section first introduces the GF-3 SAR WM dataset and European Centre for Medium-Range Weather Forecasts Reanalysis v5 (ERA-5) data used in this article, as well as the data preprocessing process, and then introduces the basic principle of polarization basis transformation and cutoff wavelength retrieval.

A. Dataset

1) WM Data of the GF-3 Satellite: The GF-3 satellite is China's first C-band (i.e., radar frequency of 5.3 GHz) multipolarization SAR satellite, launched in August 2016. It supports a total of 12 imaging modes, including Spotlight, Stripmap, ScanSAR, extended incidence angle, global observation imaging, and WM. In this study, we use the GF-3 WM data, and their parameters are described in Table I.

There are two main reasons for choosing WM data in this study. First, the quad-polarization mode of WM data meets the need of this study to obtain different polarization states. Second, the radar parameters of WM data are suitable for ocean observation, and the spatial scale is suitable for oceanographic research. The accuracy of the wind and wave parameters retrieved from the GF-3 SAR WM data has been verified [27]. In contrast to WAVEWATCH3 forecast data, the root mean square error (RMSE) of SWH retrieval results is about 0.5–0.6 m, and the mean absolute percentage error (MAPE) is about 20% [27].

In this study, we used Level 1A single-look complex WM data with a total of 4648 scenes. Among them, 2342 images were used as data for preliminary experiments to explore the influence of polarization basis on the retrieval of wind and wave parameters using the correlation between the cutoff wavelength and these parameters under different polarization bases. Further 2306 images were used as input data to retrieve the wind and wave parameters under any polarization bases. The accuracy of the retrieval of wind and wave parameters is used to demonstrate the importance of the choice of polarization basis. The geographical density distribution of these data is shown in Fig. 1.

2) ERA-5 Data: In this study, ERA-5 data provided by the European Centre for Medium-Range Weather Forecasts (ECMWF) were selected as the experimental wind and wave parameters. ERA-5 data adopt a 0.125° × 0.125° grid and be applied to forecast atmospheric, terrestrial, and marine climate conditions with a spatial resolution of approximately 30 km and a temporal resolution of 1 h. Because the ERA-5 forecast data can achieve global spatial coverage, it is well matched with the GF-3 SAR WM data.

In this study, we selected seven kinds of wind and wave parameters provided by ERA-5, including SWH, mean wave period (MWP), WS, SWH of wind waves [SWH (wind)], MWP of wind waves [MWP (wind)], SWH of the swell [SWH (swell)], and MWP of swell [MWP (swell)]. These parameters help us analyze the influence of polarization basis on the retrieval of the mixed wave, wind wave, and swell parameters.

3) Data Preprocessing: The WM data were preprocessed using radiation calibration, quality control, and data matching.

1) To facilitate later data processing, it is necessary to extract the Sinclair (S) matrix from the GF-3 WM data, which can be done by using pixel information expert (PIE)-SAR software (https://engine.piesat.cn/).
2) In order to eliminate the interference of other factors (such as ice, upwelling, and vortices) on wave observation, the quality of the images was controlled. The variance of statistical information (cvar) of all SAR images was calculated using the following equations:

$$\mu = \left\langle |I_c|^2 \right\rangle \tag{1}$$

$$\text{cvar} = \left\langle \left(|I_c|^2 - \mu \right) \right\rangle \tag{2}$$

where I_c denotes the complex data of the backscattering coefficient. In this study, images with a cvar between 1.1 and 1.6 were regarded as sufficient in quality [27]. The distribution of the cvar results is shown in Fig. 2, where the proportion of usable data is approximately 90%.
3) Because the swath of the GF-3 SAR WM data is smaller than that of the ERA-5 data, a space–time bilinear interpolation of the ERA-5 data was used to match with the GF-3 data. The equation for spatial bilinear

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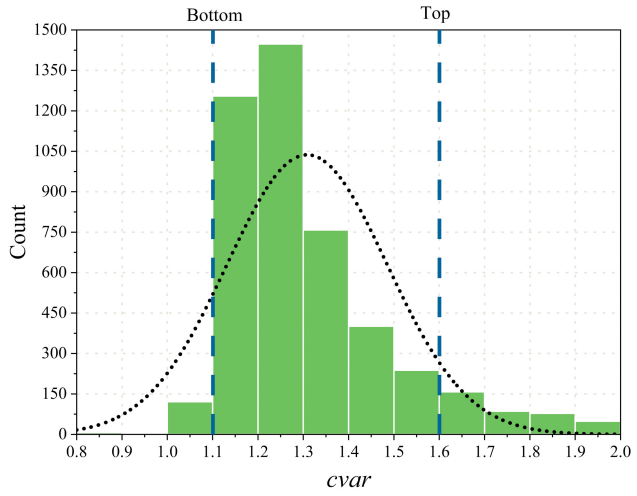

Fig. 2. Variance distribution of GF-3 SAR WM data.

interpolation is given as follows:

$$G = 64 \times (P_A \cdot d_3 \cdot d_4 + P_B \cdot d_1 \cdot d_3 + P_C \cdot d_2 \cdot d_4 + P_D \cdot d_1 \cdot d_2) \quad (3)$$

where G is the result of spatial interpolation, P_i $(i = A, B, C, D)$ is the wind and wave parameters of the four vertices in the ERA-5 grid, and d_i $(i = 1, 2, 3, 4)$ is the difference between the latitude and longitude of the four vertices of the grid and the center of the GF-3 image. A coefficient of 64 is used because a window of 0.125 is used for interpolation, and a 1-h time window is used for time interpolation. After quality filtering and space–time interpolation, 2032 scenes' GF-3 data were used to analyze the correlation between the cutoff wavelength, and wind and wave parameters under different polarization bases. A total of 2061 scene data were used to retrieve wind and wave parameters.

B. Method

1) Polarization Basis Transformation: According to the principle of electromagnetic waves [26], the Jones vector form of an elliptical polarization monochromatic plane wave under a Cartesian basis is

$$\widetilde{E} = A \begin{bmatrix} \cos\psi & -\sin\psi \\ \sin\psi & \cos\psi \end{bmatrix} \cdot \begin{bmatrix} \cos\chi & j\sin\chi \\ j\sin\chi & \cos\chi \end{bmatrix} \begin{bmatrix} e^{j\delta} & 0 \\ 0 & e^{-j\delta} \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad (4)$$

$$\widetilde{E} = AU_2(\psi)U_2(\chi)U_2(\delta)\hat{x} = AU_2(\psi, \chi, \delta)\hat{x} \quad (5)$$

where ψ is the orientation angle and is the inclination of the major axis of the polarization ellipse and χ is the elliptical angle and is a measure of the shape of the polarization ellipse. The ranges of the abovementioned parameters are $\psi \in [-(\pi/2), (\pi/2)]$ and $\chi \in [-(\pi/4), (\pi/4)]$. δ is the absolute phase term, whose value is determined by ψ and χ. Once ψ and χ are determined, δ is a constant; hence, it is not considered in general studies. $\hat{x}$ represents the unit Jones vector under horizontal polarization.

According to complex vector algebra, any complex vector can be decomposed into the sum of two orthogonal complex vectors, and hence, any one polarization base can be formed by two orthogonal polarization units' vectors. If the scalar product of the two vectors is zero, they are orthogonal. According to (5), the orthogonal Jones vector is

$$\tilde{E}_\perp = AU_2(\psi, \chi)\hat{y} = AU_2(\psi_\perp, \chi_\perp)\hat{x} \quad (6)$$

where $\psi_{\perp=}\psi + (\pi/2)$, $X_{\perp=} - X$, and $\hat{y}$ represents the unit Jones vector under vertical polarization. In addition, $\tilde{E}$ and $\tilde{E}_\perp$ constitute a pair of a Cartesian orthogonal basis. We can obtain the polarization response of any other orthogonal basis by transforming the Cartesian orthogonal basis, which is equivalent to the polarization effect in any polarization state. The electric field $\tilde{E}_{(\hat{u},\hat{u}_\perp)}$ under any polarization bases can be written as

$$\widetilde{E}_{(\hat{u},\hat{u}_\perp)} = E_u\hat{u} + E_{u_\perp}\hat{u}_\perp. \quad (7)$$

The Jones vector of the electric field relation of $\tilde{E}_{(\hat{u},\hat{u}_\perp)}$ under the Cartesian basis is

$$\begin{bmatrix} E_u \\ E_{u_\perp} \end{bmatrix} = U_2(\psi, \chi)^{-1} \begin{bmatrix} E_1 \\ E_2 \end{bmatrix}. \quad (8)$$

The Jones vector of the electric field under any polarization bases can be obtained by a special unitary matrix U_2. The S matrix obtained by polarimetric SAR is related to the incident electromagnetic wave E_i and scattering electromagnetic wave E_s as follows:

$$E_{s(1,2)} = \begin{bmatrix} S_{11} & S_{21}{}^* \\ S_{21} & S_{22} \end{bmatrix} \cdot E_{i(1,2)}. \quad (9)$$

Combining (8) and (9), we obtain

$$E_{s(u,u_\perp)} = U_2(\psi, \chi)^{-1} \cdot \begin{bmatrix} S_{11} & S_{21}{}^* \\ S_{21} & S_{22} \end{bmatrix} U_2(\psi, \chi) \cdot E_{i(u,u_\perp)} \quad (10)$$

where S_{11}, S_{21}, and S_{22} ($*$ is a conjugate operation) denote the complex scattering coefficients of HH, VH, and VV under H-V Cartesian linear polarization, respectively. The polarization basis transformation of scattering matrix S can be obtained as follows:

$$\begin{bmatrix} S_{uu} & S_{uu_\perp} \\ S_{u_\perp u} & S_{u_\perp u_\perp} \end{bmatrix} = U_2(\psi, \chi)^{-1} \cdot \begin{bmatrix} S_{11} & S_{21}{}^* \\ S_{21} & S_{22} \end{bmatrix} U_2(\psi, \chi) \quad (11)$$

where S matrix (which can be directly obtained from the quad-polarization SAR data) of the H-V polarization base can be transformed into the polarization state under any polarization bases by setting ψ and χ.

Taking the transformation of an H-V polarization base to a left circular–right circular (LR) polarization base as an example, polarization basis transformation requires a set of orthogonal polarization basis as the prerequisite. Instead of using the LR polarization base, we use the left circular–left orthogonal circular polarization base even if the left orthogonal polarization corresponds to the right one. Under a left circular-left orthogonal circular polarization base, the following two special unitary matrix transformations can be given:

$$U_2(\psi = 0°, \chi = 45°)^{-1} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -j \\ -j & 1 \end{bmatrix} \quad (12)$$

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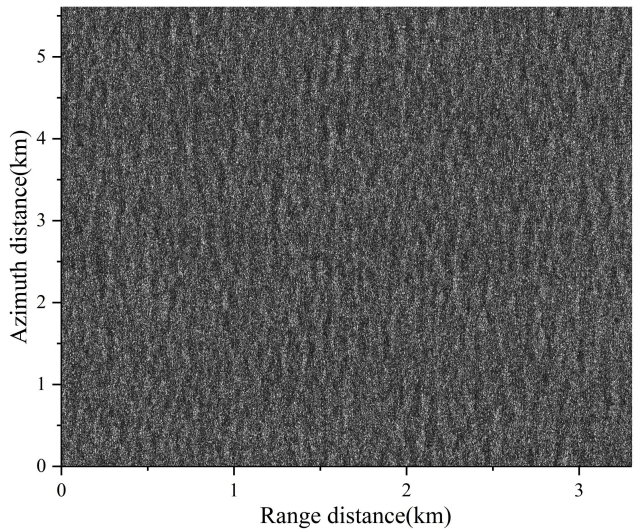

Fig. 3. GF-3 SAR WM image. Time: 15:19:43 UTC, January 29, 2017; polarization: VV; center longitude: −136.38 °W; center latitude: 39.25 °N; center incidence angle: 23.62°; NRCS: −7.20 dB; and normalized variance: 1.42.

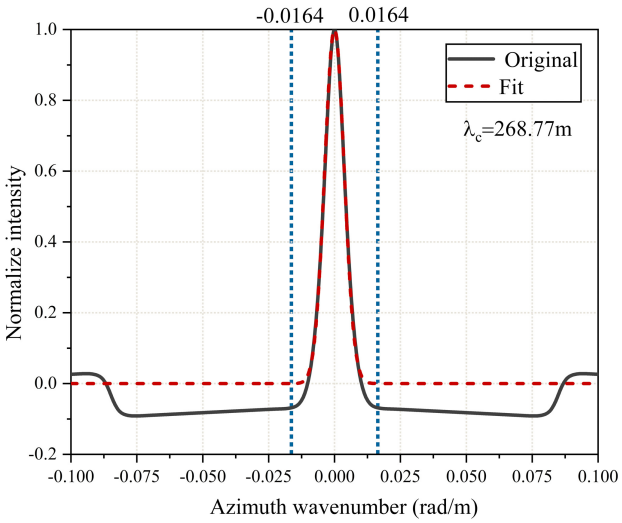

Fig. 4. Retrieval results of the cutoff wavelength of GF-3 SAR WM (black solid line is the normalized intensity value of the original data, red dashed line is the fitting curve obtained by Gaussian fitting, and blue dashed line is the cutoff wavenumber corresponding to the cutoff wavelength).

$$U_2(\psi = 0°, \chi = 45°) = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & j \\ j & 1 \end{bmatrix}. \tag{13}$$

Substituting (12) and (13) into (11), the scattering matrix of the LR polarization base can be expressed as follows:

$$\begin{bmatrix} S_{11}(\text{Cir}) & S_{21}{}^*(\text{Cir}) \\ S_{21}(\text{Cir}) & S_{22}(\text{Cir}) \end{bmatrix} = \frac{1}{2} \cdot \begin{bmatrix} S_{11} + S_{22} + (S_{21}{}^* - S_{21})j & S_{21}{}^* + S_{21} + (S_{11} - S_{22})j \\ S_{21}{}^* + S_{21} + (S_{22} - S_{11})j & S_{11} + S_{22} + (S_{21} - S_{21}{}^*)j \end{bmatrix} \tag{14}$$

where S_{11} (Cir), S_{21} (Cir), and S_{22} (Cir), respectively, denote that S_{11}, S_{21}, and S_{22} have been converted into a circular polarization base by a polarization basis transformation. In this way, we can change the geometric parameters ψ and χ of the polarization ellipse to obtain the polarization response S_{ij} (ψ, χ) under any other basis, which is equivalent to obtain the target scattering information under this polarization basis.

2) Cutoff Wavelength Retrieval Based on the SAR Cross Spectrum: A common empirical method to retrieve the azimuth cutoff wavelength is to use the cross correlation of the SAR cross-spectrum fitting with a Gaussian function [3]. This method can improve the signal-to-noise ratio of the image and suppress the contribution of nonpolarized scattering to a certain extent [3], [28].

Taking the SAR image of Fig. 3 as an example, the specific process of cutoff wavelength retrieval based on cross spectrum is given as follows.

1) The SAR image is transformed from the spatial domain to the range-Doppler (RD) domain by azimuth fast Fourier transform (FFT). The mean profile of the RD domain shows a *sinc* window function, and the window function must be removed before we can create individual sublooks. Removal of the signal is done by fitting a *cosine* curve to the azimuth profile using a least-squares optimization. The magnitude of the original profile is divided by the magnitude of the fit profile, resulting in a rectangular profile in the RD domain.
2) The rectangular section is divided into three sublooks, and the cross spectrum of sublooks I and III is calculated. The equation is given as follows:

$$P_s^{(1,3)}(k, \Delta t) = \frac{1}{\langle I^{(1)} \rangle \cdot \langle I^{(3)} \rangle} \cdot \left\langle I^{(1)}\left(k, \frac{t}{2}\right) \cdot I^{*(3)}\left(k, -\frac{t}{2}\right) \right\rangle \tag{15}$$

where $\langle I^{(m)} \rangle$ is the average image intensity, $I^{(m)}(k, (t/2))$ is the Fourier transform of the intensity map, and $\Delta t = (t_m/2) - (-(t_n/2))$ is the sublook separation time between sublook intensity images. In this article, the reason for choosing sublooks I and III is that the time interval between adjacent sublooks is relatively small, which will lead to small changes in the phase of the wave in the limited integral time and, thus, cause relatively large levels of noise, making it difficult to obtain a high-quality cross spectrum [29]. According to the imaging time of the GF-3 WM, the time interval between sublooks I and III is approximately 0.56 s.
3) A Fourier transform is performed on the real part of the cross spectrum, and the autocorrelation function in the azimuth direction is obtained. A Gaussian function is used to fit the autocorrelation function as follows:

$$C(x) \sim e^{-\pi^2 \frac{x^2}{\lambda^2}}. \tag{16}$$

The cutoff wavenumber can be obtained by the value of the fitting coefficient λ, and then, the cutoff wavelength λ_c can be obtained, as shown in Fig. 4.

III. Correlation Analysis Under Different Polarization Base

In this section, the correlation between the cutoff wavelength, and the wind and wave parameters under special

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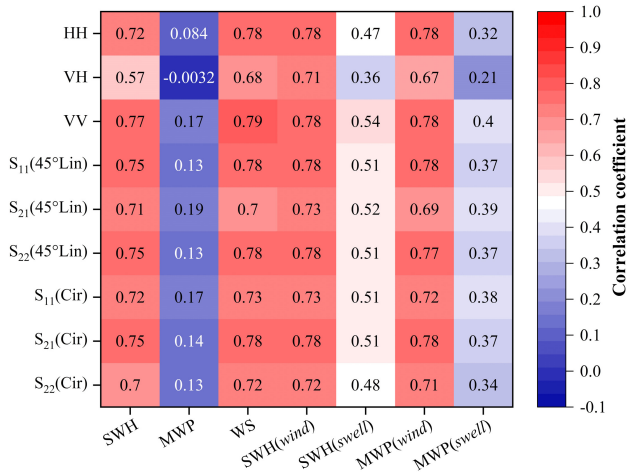

Fig. 5. COR diagram of cutoff wavelength with wind and wave parameters in the special polarization state.

polarization bases (the H-V linear, circular, and 45° linear rotated polarization bases) is evaluated, and then, the analysis case is extended to the general elliptical polarization bases. Finally, the analysis results are used as a theoretical reference for the retrieval of wind and wave parameters.

A. Correlation of the Cutoff Wavelength With Wind and Wave Parameters Under Special Polarization Bases

The special polarization bases considered in this section are the H-V linear, circular, and 45° linear rotated polarization bases. As shown in Fig. 5, the correlation between the cutoff wavelength, and wind and wave parameters under each special polarization basis was evaluated. The numbers in the colored cells represent the Pearson correlation coefficient (COR) results of the 2032 GF-3 WM data under different polarization bases. The COR is a statistic reflecting the degree of linear correlation of two variables. A larger COR indicates a stronger correlation between the two variables, which is helpful for the simplification of the retrieval model and improving the accuracy of the retrieval results.

In Fig. 5, the axis indicates the polarization state under different polarization bases. HH, VH, and VV represent the initial polarization states of quad-polarization data. S_{ij}(45°Lin) represents the complex scattering coefficient converted into a 45° linear rotated polarization base state through the polarization basis transformation; S_{ij}(Cir) represents the complex scattering coefficient converted into LR circular polarization base state through the polarization basis transformation. The transverse axis represents different wind and wave parameters. (Because of the reciprocity between VH and HV, only VH is considered in this study.)

Fig. 5 reveals that different polarization bases influence the correlation between the cutoff wavelength and the wind and wave parameters. In the quad-polarization states, the correlation between VH polarization and other polarization states is substantially lower than those of other polarization states. This occurs for two reasons. The first reason comes from the inherent background noise of SAR. VH polarization is seriously affected by background noise, which reduces the signal-to-noise ratio of the VH polarization state. The second reason is that VH polarization is more vulnerable to the influence of nonpolarized scattering. The main contribution of nonpolarized scattering comes from breaking waves, and rough patches with very high local acceleration and a short life greatly affect the observation of waves using VH polarization. SAR cannot deal with these patches within a limited integration time, which results in low azimuth resolution of the VH polarization. Therefore, in contrast to the polarized radar echo, the nonpolarized radar echo controlled by the foam of a breaking wave is a part of the factor that leads to a large difference between the polarization states. Bragg wave modulation with a small relaxation scale can be considered limited under this situation.

The measured cutoff wavelength of HH polarization is larger than that of VV polarization, which causes the difference between HH polarization and VV polarization. The main reason is that the HH signal is also sensitive to nonpolarized scattering, and the contribution of a breaking wave can reach 60% to 70% of the total backscattering contribution under HH polarization through specular reflection [30]. In contrast to VV polarization observation, HH polarization observation usually minimizes the polarized contribution of small-scale Bragg waves, which makes HH polarization more sensitive to random motion. In addition, in VV polarization, the radar signal is more sensitive to smaller waves than in HH polarization, and HH polarization leads to a larger Doppler frequency at high incident angles because of the influence of wave phase velocity [31]. In summary, nonpolarized scattering plays a significant role in the significant difference between HH and VV, and we also speculate that the contribution of background noise on VH polarization may be greater.

In addition, in Fig. 5, the correlation between the cutoff wavelength and the wind and wave parameters under 45° linear rotated polarization and circular polarization is generally worse than that of VV polarization. The results reveal that for different wind and wave parameters, the correlation with the cutoff wavelength varies substantially. Among them, the CORs of SWH, WS, SWH (wind), and MWP (wind) are generally higher than 0.7. Those of SWH (swell) and MWP (swell) fluctuate around 0.5 and 0.4, respectively. There is almost no correlation between MWP and the cutoff wavelength. The above discussion reveals that the cutoff wavelength has a good correlation with mixed wave and wind wave parameters but has a poor correlation with swell parameters. This is consistent with our understanding that the cutoff wavelength is closely related to the high-frequency waves, and it mainly affects the wind wave part with high frequency in the wave spectrum retrieval.

B. Correlation of the Cutoff Wavelength With Wind and Wave Parameters Under Elliptical Polarization Bases

In this section, the correlation between the cutoff wavelength and the wind and wave parameters under an elliptical polarization base is analyzed. In addition, the polarization modes suitable for various wind and wave parameters are determined based on the analysis results.

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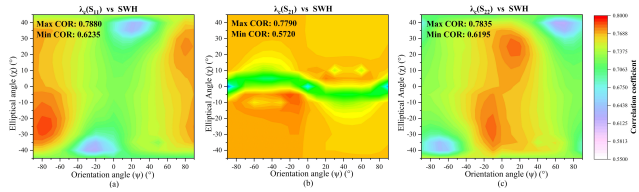

Fig. 6. COR between cutoff wavelength and SWH under general elliptical polarization base and polarization basis transformation based on (a) S_{11}, (b) S_{21}, and (c) S_{22}.

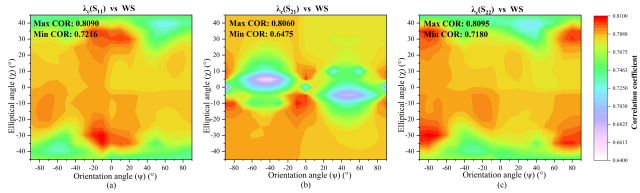

Fig. 7. COR between cutoff wavelength and WS under general elliptical polarization base and polarization basis transformation based on (a) S_{11}, (b) S_{21}, and (c) S_{22}.

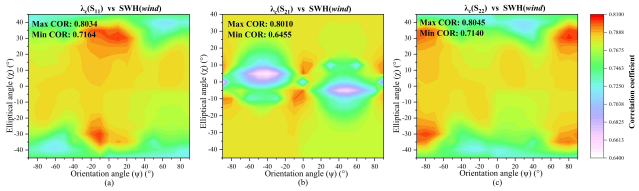

Fig. 8. COR between cutoff wavelength and SWH (wind) under general elliptical polarization base and polarization basis transformation based on (a) S_{11}, (b) S_{21}, and (c) S_{22}.

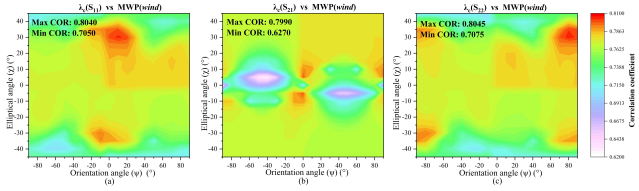

Fig. 9. COR between cutoff wavelength and MWP (wind) under general elliptical polarization base and polarization basis transformation based on (a) S_{11}, (b) S_{21}, and (c) S_{22}.

Using the polarization basis transformation described above, we can obtain the cutoff wavelength of S_{ij} (ψ, χ) under an elliptical polarization base by changing the polarization ellipse parameters ψ and χ. Figs. 6–9 present diagrams of the COR between the cutoff wavelength, and various wind and wave parameters under different polarization bases. The wind and wave parameters are SWH, WS, SWH (wind), and MWP (wind). The reason for choosing these parameters is that they show a strong correlation with the cutoff wavelength under the three special polarization bases. Please note that the angle intervals of ψ and χ in this study are 10° and 5°, respectively, and we obtain the contours as shown in Figs. 6–10 by interpolating the logarithmic values.

Fig. 6 illustrates the COR between the cutoff wavelength and SWH. Fig. 6(a)–(c), respectively, shows the performance of S_{11}, S_{21}, and S_{22} under different polarization bases. The results indicate the following.

1) The intensity of the COR is distributed symmetrically, and larger COR values are concentrated in the lower left and upper right corners of Fig. 6(a) and the central area of the upper and lower parts of Fig. 6(c), which fully proves that the sensitivity of wind and wave parameters strongly depend on radar polarization. In addition, as shown in Fig. 6, according to the principle of the polarization basis transformation, it is obvious that HH corresponds to $S_{11}(0°, 0°)$, and VV is equivalent to $S_{11}(90°, 0°)$. Thus, increases or decreases between $S_{11}(90°, 0°)$ and $S_{11}(0°, 0°)$ with ψ in Fig. 6 are actually the process of HH polarization changing to VV polarization. By analogy, the increase or decrease in $S_{22}(0°, 0°)$ with ψ in Fig. 6(c) is actually the process of VV polarization changing to HH polarization, which explains why Fig. 6(a) and (c) shows quite similar contour.
2) In all polarization bases, the maximum value of the COR is 0.788, which is slightly better than the 0.767 of VV polarization. Moreover, the minimum value of 0.572 is obtained for cross polarization, and the values of other cross polarizations are also below 0.6. As we expected, cross polarization is more affected by other factors than the other polarization states.

Figs. 7–9 show the COR between the cutoff wavelength, SWH, WS, SWH (wind), and MWP (wind), respectively. These parameters are all directly related to the sea surface wind. Therefore, we analyze these all wind-driven parameters together. The following can be inferred from our results.

1) Strong correlation values are concentrated in the four corners of Figs. 7(c), 8(c), and 9(c) and the upper and lower central regions in Figs. 7(a), 8(a), and 9(a). The COR distributions of wind-related parameters in Figs. 7–9 are generally consistent, but the distributions are significantly different from those in Fig. 6. Therefore, swell also influences SWH's polarization sensitivity, and this cannot be ignored.
2) The maximum value of the COR of the three wind-driven parameters can exceed 0.8, which is higher than that of VV polarization.

To better explain why the performance of the COR between mixed waves and wind waves is quite different, the COR between cutoff wavelength and SWH (swell) is analyzed as shown in Fig. 10. We can see that the pattern of COR distribution of mixed wave (see Fig. 6) is similar to swell wave

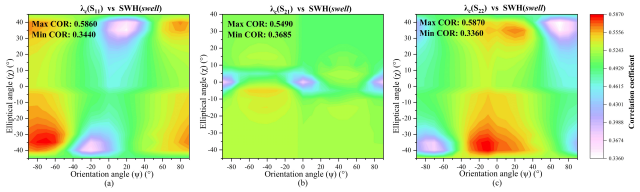

Fig. 10. COR between cutoff wavelength and SWH (swell) under general elliptical polarization base and polarization basis transformation based on (a) S_{11}, (b) S_{21}, and (c) S_{22}.

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(see Fig. 10) but quite different from wind wave (see Figs. 7–9). It suggests that the polarization sensitivity of mixed waves is strongly affected by swell effects in SAR data. The reason for the result is that wind waves dominated by high-frequency waves are highly affected by random shifts, resulting in a destructive wavy pattern known as azimuth cutoff. Thus, the wind wave information in SAR imagettes will be lost and show relatively clear wavy patterns for swell [32], [33], [34].

Overall, there are many elliptical polarization bases that are more sensitive to wind and wave parameters than the H-V polarization base, as seen in Figs. 6–10. In addition, the polarization ellipse parameters have little effect on the COR under the circular polarization bases. Regardless of whether the orientation angle and elliptical angle change, the circular polarization bases have a medium value of CORs.

IV. Theoretical Explanation of the Correlation Analysis Results

As mentioned before, there are many elliptical polarization bases that have higher CORs than H-V linear, circular, and linear rotated polarization bases. This section will explain the reason for this.

According to Hajnsek *et al.* [35], the polarization coherence matrix of electromagnetic waves is generally expressed as

$$T_3 = \begin{bmatrix} t_{11} & t_{12} & t_{13} \\ t_{12}^* & t_{22} & t_{23} \\ t_{13}^* & t_{23}^* & t_{33} \end{bmatrix} \tag{17}$$

and the Stokes vector of (17) can be written as follows, according to [36]:

$$\underline{g}_s = \begin{bmatrix} g_0 \\ g_1 \\ g_2 \\ g_3 \end{bmatrix} = \begin{bmatrix} \frac{1}{2}(t_{11} + t_{22} + t_{33}) \pm \mathrm{Im}(t_{23}) \\ \mathrm{Re}(t_{12}) \pm \mathrm{Im}(t_{13}) \\ \mathrm{Re}(t_{13}) \pm \mathrm{Im}(t_{12}) \\ \mathrm{Im}(t_{23}) \pm \frac{1}{2}(t_{22} + t_{33} - t_{11}) \end{bmatrix}. \tag{18}$$

For the Bragg scattering of rough surfaces, the scattering model satisfies the rank-1 symmetric scattering mechanism [37]. The scattering coherence matrix can be written as

$$T_3 = m_s \begin{bmatrix} \cos^2 \alpha_s & \cos \alpha_s \sin \alpha_s e^{i\phi} & 0 \\ \cos \alpha_s \sin \alpha_s e^{-i\phi} & \sin^2 \alpha_s & 0 \\ 0 & 0 & 0 \end{bmatrix}. \tag{19}$$

Among them, m_s is the total polarization power, which contains the information on small-scale sea surface roughness [38]. The scattering angle α_s depends on the local incidence angle and is closely related to the ellipticity of the scattering wave. The additional term ϕ contains the phase information of the scattering wave. The Stokes vector of (19) can be written as

$$\underline{g}_s = \frac{m_s}{2} \begin{bmatrix} 1 \\ \sin 2\alpha_s \cos \phi \\ \pm \sin 2\alpha_s \sin \phi \\ \pm \cos 2\alpha_s \end{bmatrix}. \tag{20}$$

For Bragg scattering of untilted rough surface, the maximum copolarization signature occurs at an $\psi = 90°$ and $\chi = 0°$ (i.e., at VV polarization). However, a tilted rough surface in the azimuth direction introduces HV and VH components, and the depolarization effect, which causes the shift of the location of the maximum response in the copolarization signature away from $\psi = 90°$. The amount of shift from $\psi = 90°$ relates to the slope of the tilted surface. For the sea surface scattering, the radar's scattering of Bragg waves is modulated by the long wave, which is commonly represented by the two-scale Bragg scattering model [39]. In the two-scale Bragg scattering model, the effect of the tilt surface is due to the modulation of the long wave on the Bragg wave. A way to introduce a tilted surface in the azimuth direction by rotating the Bragg coherency matrix (19) about an angle in the plane perpendicular to the scattering plane, as shown in Fig. 11 [35]

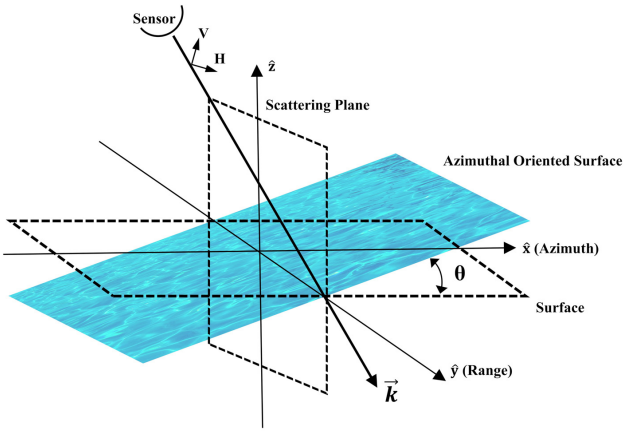

Fig. 11. In the two-scale Bragg scattering model, the modulation of long waves on Bragg waves makes the sea surface rotate θ degrees around the line of sight.

$$T3\ R = 2R(\theta) \begin{bmatrix} t_{11} & t_{12} & 0 \\ t_{12}^* & t_{22} & 0 \\ 0 & 0 & 0 \end{bmatrix} R(-\theta) \tag{21}$$

where $R(\theta)$ is the rotation matrix, representing the effect of the Bragg coherence matrix being rotated θ in a plane perpendicular to the scattering plane

$$R(\theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos 2\theta & -\sin 2\theta \\ 0 & \sin 2\theta & \cos 2\theta \end{bmatrix}. \tag{22}$$

Substituting (19) and (22) into (21), the two-scale Bragg scattering model can be obtained

$$T3R = m_s \begin{bmatrix} 2\cos^2 \alpha_s & \cos 2\theta \sin 2\alpha_s e^{i\phi} & \sin 2\theta \sin 2\alpha_s e^{i\phi} \\ \cos 2\theta \sin 2\alpha_s e^{-i\phi} & 2\cos^2 2\theta \sin^2 \alpha_s & \sin 4\theta \sin^2 \alpha_s \\ \sin 2\theta \sin 2\alpha_s e^{-i\phi} & \sin 4\theta \sin^2 \alpha_s & 2\sin^2 2\theta \sin^2 \alpha_s \end{bmatrix}. \tag{23}$$

Its Stokes vector can be written as

$$\underline{g}_s = m_s \begin{bmatrix} 1 \\ \cos 2\theta \sin 2\alpha_s \mathrm{Re}\left(e^{i\phi}\right) \pm \sin 2\theta \sin 2\alpha_s \mathrm{Im}\left(e^{i\phi}\right) \\ \sin 2\theta \sin 2\alpha_s \mathrm{Re}\left(e^{i\phi}\right) \pm \cos 2\theta \sin 2\alpha_s \mathrm{Im}\left(e^{i\phi}\right) \\ \pm\left(\sin^2 \alpha_s - \cos^2 \alpha_s\right) \end{bmatrix}. \tag{24}$$

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In the radar footprint range, the tilt angle is not a constant value and follows a certain probability distribution $\rho(\theta)$. The coherence matrix can be calculated by performing a configurational averaging over a given distribution $\rho(\theta)$ of θ

$$[T] = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} T(\theta)\rho(\theta)d\theta. \tag{25}$$

Then, it can be obtained as follows:

$$[T3\ R] = m_s \begin{bmatrix} 2\cos^2\alpha_s & X_{12}\sin 2\alpha_s e^{i\phi} & 0 \\ X_{12}\sin 2\alpha_s e^{-i\phi} & 2\ X_{22}\sin^2\alpha_s & 0 \\ 0 & 0 & 2\ X_{33}\sin^2\alpha_s \end{bmatrix}. \tag{26}$$

The model parameters X_{12}, X_{22}, and X_{33} are given as follows:

$$\begin{aligned} X_{12} &= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos(2\theta)\rho(\theta)d\theta \\ X_{22} &= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos^2(2\theta)\rho(\theta)d\theta \\ X_{33} &= 1 - X_{22}. \end{aligned} \tag{27}$$

The Stokes vector of (26) can be written as

$$\left[\underline{g}_s\right] = m_s \begin{bmatrix} 1 \\ X_{12}\sin 2\alpha_s \operatorname{Re}\left(e^{i\phi}\right) \\ \pm X_{12}\sin 2\alpha_s \operatorname{Im}\left(e^{i\phi}\right) \\ \pm[(X_{22}+X_{33})\sin^2\alpha_s - \cos^2\alpha_s] \end{bmatrix}. \tag{28}$$

In the following, we will discuss the expression of the two-scale Bragg scattering model under three common distributions (see Fig. 12). The probability density function and the corresponding model parameters of the uniform distribution, the half cosine distribution, and the Lee distribution [40], [41], [42] are shown in (29)–(31), shown at the bottom of the page.

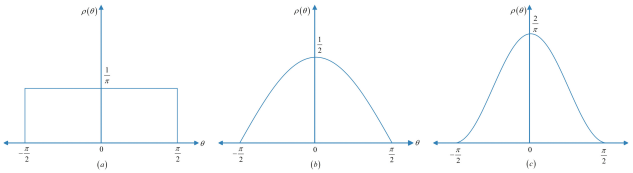

Fig. 12. Three statistical distributions for the two-scale Bragg scattering model: (a) uniform, (b) half cosine with $\Delta = \pi$, (c) and Lee with $\Delta = \pi$.

$$\rho(\theta) = \begin{cases} \dfrac{1}{\Delta}, & -\dfrac{\Delta}{2} \le \theta \le \frac{\Delta}{2} \\ 0, & \text{elsewhere} \end{cases}$$

$$\begin{cases} X_{12}^U = \sin c(\Delta) \\ X_{22}^U = 0.5(1+\sin c(2\Delta)) \\ X_{33}^U = 1 - X_{22}^U \end{cases} \tag{29}$$

where the distribution range of Δ is from 0° to 180°. Through the above expression, the Stokes vectors of the two-scale Bragg scattering model under the uniform distribution, the half cosine distribution, and the Lee distribution are shown in (32)–(34), shown at the bottom of the page.

$$\left[\underline{g}_s\right] = m_s \begin{bmatrix} 1 \\ \sin c(\Delta)\sin 2\alpha_s \operatorname{Re}\left(e^{i\phi}\right) \\ \pm\sin c(\Delta)\sin 2\alpha_s \operatorname{Im}\left(e^{i\phi}\right) \\ \pm\left(\sin^2\alpha_s - \cos^2\alpha_s\right) \end{bmatrix}. \tag{32}$$

$$\rho(\theta) = \begin{cases} \frac{\pi}{2\Delta}\cos\frac{\pi\theta}{\Delta}, & -\frac{\Delta}{2} \le \theta \le \frac{\Delta}{2} \\ 0, & \text{elsewhere} \end{cases}$$

$$\begin{cases} X_{12}^H = \frac{\pi}{2\Delta}\left[\frac{\Delta}{\pi+2\Delta}\sin\left(\frac{\pi+2\Delta}{2}\right) + \frac{\Delta}{\pi-2\Delta}\sin\left(\frac{\pi-2\Delta}{2}\right)\right] \\ X_{22}^H = \frac{1}{2}\sin\left(\frac{\pi}{2}\right) + \frac{\pi}{4\Delta}\left[\frac{\Delta}{\pi+4\Delta}\sin\left(\frac{\pi+4\Delta}{2}\right) + \frac{\Delta}{\pi-4\Delta}\sin\left(\frac{\pi-4\Delta}{2}\right)\right] \\ X_{33}^H = 1 - X_{22}^H \end{cases} \tag{30}$$

$$\rho(\theta) = \begin{cases} \frac{1}{\Delta}\left(1+\cos\frac{2\pi}{\Delta}\theta\right), & -\frac{\Delta}{2} \le \theta \le \frac{\Delta}{2} \\ 0, & \text{elsewhere} \end{cases}$$

$$\begin{cases} X_{12}^L = \sin c(\Delta) - \frac{\Delta}{2\pi+2\Delta}\sin c(\Delta) + \frac{\Delta}{2\pi-2\Delta}\sin c(\Delta) \\ X_{22}^L = \frac{1}{2}(1+\sin c(2\Delta)) - \frac{\Delta}{2\pi+4\Delta}(\sin c(2\Delta)) + \frac{\Delta}{2\pi-4\Delta}(\sin c(2\Delta)) \\ X_{33}^L = 1 - X_{22}^L \end{cases} \tag{31}$$

$$\left[\underline{g}_s\right] = m_s \begin{bmatrix} 1 \\ \frac{\pi}{2\Delta}\left[\frac{\Delta}{\pi+2\Delta}\sin\left(\frac{\pi+2\Delta}{2}\right) + \frac{\Delta}{\pi-2\Delta}\sin\left(\frac{\pi-2\Delta}{2}\right)\right]\sin 2\alpha_s \operatorname{Re}\left(e^{i\phi}\right) \\ \pm\frac{\pi}{2\Delta}\left[\frac{\Delta}{\pi+2\Delta}\sin\left(\frac{\pi+2\Delta}{2}\right) + \frac{\Delta}{\pi-2\Delta}\sin\left(\frac{\pi-2\Delta}{2}\right)\right]\sin 2\alpha_s \operatorname{Im}\left(e^{i\phi}\right) \\ \pm\left(\sin^2\alpha_s - \cos^2\alpha_s\right) \end{bmatrix} \tag{33}$$

$$\left[\underline{g}_s\right] = m_s \begin{bmatrix} 1 \\ \left[\sin c(\Delta) - \frac{\Delta}{2\pi+2\Delta}\sin c(\Delta) + \frac{\Delta}{2\pi-2\Delta}\sin c(\Delta)\right]\sin 2\alpha_s \operatorname{Re}\left(e^{i\phi}\right) \\ \pm\left[\sin c(\Delta) - \frac{\Delta}{2\pi+2\Delta}\sin c(\Delta) + \frac{\Delta}{2\pi-2\Delta}\sin c(\Delta)\right]\sin 2\alpha_s \operatorname{Im}\left(e^{i\phi}\right) \\ \pm\left(\sin^2\alpha_s - \cos^2\alpha_s\right) \end{bmatrix} \tag{34}$$

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TABLE II
COR COMPARISON BETWEEN PEAK STATE AND VV POLARIZATION

Parameters	peak state (ψ,χ)	COR	
		COR(peak state)	COR(VV)
SWH	$S_{11}\,(-80°, -25°)$	0.788	0.769
WS	$S_{22}\,(-80°, -30°)$	0.809	0.783
SWH($wind$)	$S_{22}\,(80°, 30°)$	0.804	0.781
MWP($wind$)	$S_{22}\,(80°, 30°)$	0.804	0.779

The Stokes vector of scattering wave in common linear polarization and circular polarization can be written as

$$\underline{g}_H = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \quad \underline{g}_V = \begin{bmatrix} 1 \\ -1 \\ 0 \\ 0 \end{bmatrix}$$
$$\underline{g}_{45°} = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \quad \underline{g}_L = \begin{bmatrix} 1 \\ 0 \\ 0 \\ -1 \end{bmatrix}, \quad \underline{g}_R = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \end{bmatrix} \tag{35}$$

where $\underline{g}_H$, $\underline{g}_V$, and $\underline{g}_{45°}$, respectively, represent the Stokes vectors of the scattering wave in the horizontal, vertical, and 45° directions, and $\underline{g}_L$ and $\underline{g}_R$, respectively, represent the Stokes vectors of the scattering wave in the left circular and right circular directions.

By comparing the Stokes vectors of the above equations with those of the two-scale Bragg scattering model under three distributions, it can be seen that, under the three typical statistical characteristic distributions, the Stokes vector of the received echo is no longer consistent with the linear polarization or circular polarization because the modulation of the long wave on the short Bragg wave generates the polarization azimuth offset, which makes the cross correlation between the horizontal polarization and the vertical polarization less than 1. This means that, under the influence of a tilted surface, the sea surface echo of electromagnetic waves changes from linear polarization to elliptical polarization; hence, using the elliptical polarization bases will make the echo information more relevant to the ocean waves.

V. PARAMETER RETRIEVAL PERFORMANCE EVALUATION

To verify whether the application of elliptical polarization bases can enhance the capability of wind and wave parameter retrieval, we chose the polarization basis with the maximum CORs for each parameter and named it "peak state" in the study. Polarization ellipse parameters of the "peak state" and the COR differences between "peak state" and VV polarization are shown in Table II.

A. Comparison of Wind and Wave Parameters Retrieval Accuracy Between "Peak State" and VV Polarization

In this study, the CWAVE-ENV empirical algorithm [43] was used to retrieve wind and wave parameters. The relationship between SAR image parameters, and wind and wave parameters is constrained by the following multiple linear regression model:

$$W = a_0 + \sum_{i=1}^{M} a_i k_i + \sum_{i=1}^{M} \sum_{j=1}^{i} a_{i,j} k_i k_j \tag{36}$$

where W denotes the retrieval parameters of wind or wave, a_i is the model parameter, k_i denotes the correlated explanatory variables in the SAR image, and M represents the number of training sample data pairs. To better demonstrate the importance of different polarization bases in the retrieval of wind and wave parameters based on the cutoff wavelength, we simplify the CWAVE-ENV model. The explanatory variable parameter k_i selected by the model is mostly related to the polarization basis after the basis transformation. The four explanatory variables are λ_c, normalized radar backscattering cross section (NRCS), σ^2, and the incident angle. The first three parameters are obtained from SAR images under different polarization bases, and the results are only affected by the polarization basis. The incident angle in each model is a constant input, and hence, it does not have an additional impact on the results. The selection of wind and wave parameters, i.e., SWH, WS, SWH (wind), and MWP (wind), is consistent with the description in Section III. It should be further clarified that, to compare the retrieved wind and wave results from VV polarization, only one polarization basis corresponding to the "peak state" in Table II was selected. Taking the SWH as example, we use only the "peak state" $S_{11}(-80°, -25°)$.

In this section, 2061 scenes of GF-3 SAR WM data were used as the input of the model. The evaluation indexes of the experimental results are RMSE, mean absolute error (MAE), MAPE, and COR, which are expressed as follows:

$$\text{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(y_i - x_i)^2} \tag{37}$$

$$\text{MAE} = \frac{1}{N}\sum_{i=1}^{N}\sqrt{(y_i - x_i)^2} \tag{38}$$

$$\text{MAPE} = \frac{1}{N}\sum_{i=1}^{N}\sqrt{\left(\frac{y_i - x_i}{x_i}\right)^2} \tag{39}$$

$$\text{COR} = \frac{\sum_{i=1}^{N}(x_i - \langle x_i\rangle)(y_i - \langle y_i\rangle)}{\sqrt{\sum_{i=1}^{N}(x_i - \langle x_i\rangle)^2 \sum_{i=1}^{N}(y_i - \langle y_i\rangle)^2}} \tag{40}$$

where N is the number of verification samples, and x_i and y_i represent the wind and wave parameters obtained by the ith GF-3 retrieval and the reference data provided by the corresponding ERA-5, respectively. According to Table II, the retrieval results of the wind and wave parameters under the above polarization states are shown in Figs. 13–16.

As shown in Fig. 13, in comparison with the SWH of ERA-5, the RMSE of SWH retrieved from VV polarization is 0.63 m, while it decreases to 0.59 m when using $S_{11}(-80°, -25°)$. Moreover, the other performance errors have also been decreased. The results of Figs. 14–16 are similar to Fig. 13. After the transformation of the polarization basis, the

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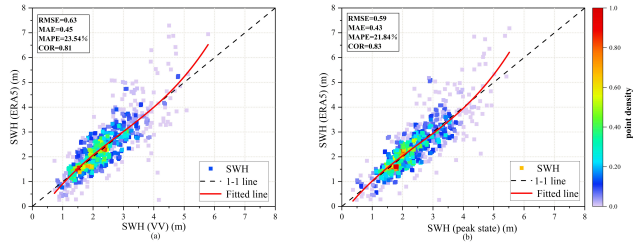

Fig. 13. Density scatter plot of (a) VV polarization and (b) $S_{11}(-80°, -25°)$ polarization retrieval SWH and ERA-5 predicted SWH.

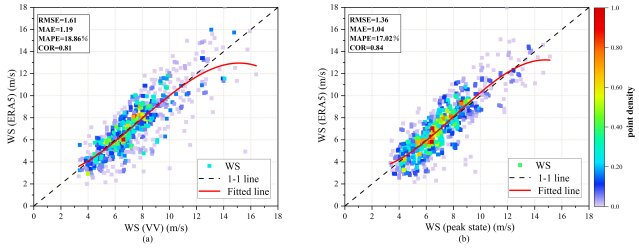

Fig. 14. Density scatter plot of (a) VV polarization and (b) $S_{22}(-80°, -30°)$ polarization retrieval WS and ERA-5 predicted WS.

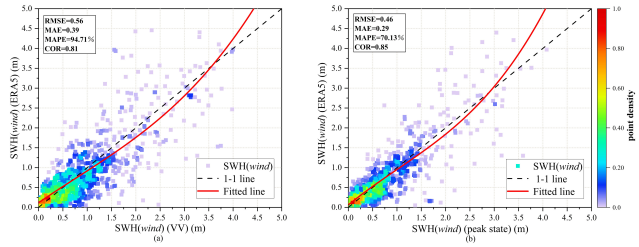

Fig. 15. Density scatter plot of (a) VV polarization and (b) $S_{22}(80°, 30°)$ polarization retrieval SWH (wind) and ERA-5 predicted SWH (wind).

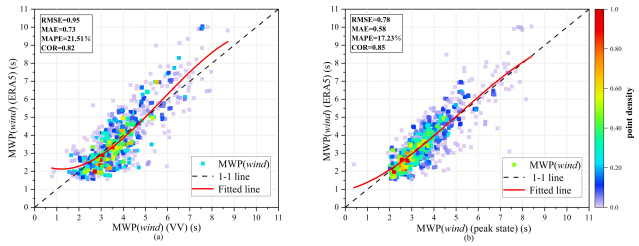

Fig. 16. Density scatter plot of (a) VV polarization and (b) $S_{22}(80°, 30°)$ polarization retrieval MWP (wind) and ERA-5 predicted MWP (wind).

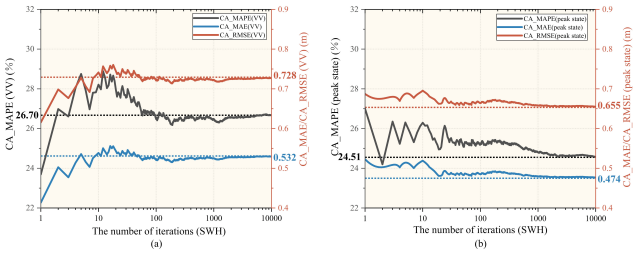

Fig. 17. Iteration diagram of accuracy index of SWH model retrieval by (a) VV polarization and (b) $S_{11}(-80°, -25°)$ polarization cutoff wavelength.

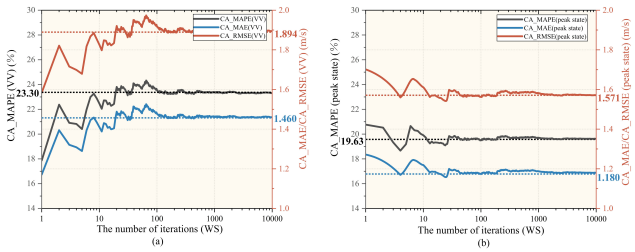

Fig. 18. Iteration diagram of accuracy index of WS model retrieval by (a) VV polarization and (b) $S_{22}(-80°, -30°)$ polarization cutoff wavelength.

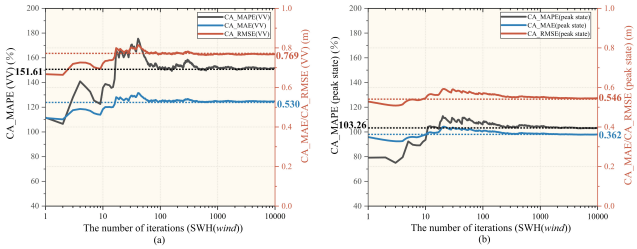

Fig. 19. Iteration diagram of accuracy index of SWH (wind) model retrieval by (a) VV polarization and (b) $S_{22}(80°, 30°)$ polarization cutoff wavelength.

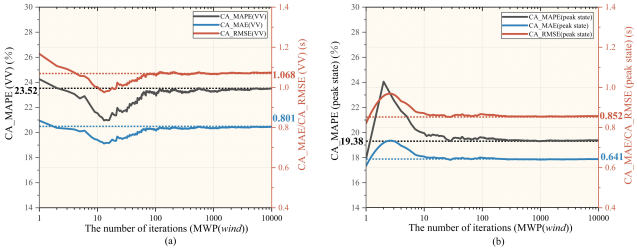

Fig. 20. Iteration diagram of accuracy index of MWP (wind) model retrieval by (a) VV polarization and (b) $S_{22}(80°, 30°)$ polarization cutoff wavelength.

ability of the cutoff wavelength to retrieve wind and wave parameters has been improved. It should be noted that the SWH retrieval in Fig. 15 is seriously overestimated. This is because approximately 45% of SWH (wind) data provided by ERA-5 is less than 0.5 m, which results in an imbalance in the number of statistical samples.

Only conducting the experiment once would result in highly randomized results in the comparison experiment. In order to solve this problem, we adopted the cumulative average (CA) method, which selected samples repeatedly and randomly as training and validation data for samples' expansion [44]. The CA method is calculated as follows:

$$A_n = \frac{1}{n}\sum_{m=1}^{n} P_m \tag{41}$$

where A_n is the average value of the n iterations and P_m is the model accuracy of the mth iteration. After a large amount of training, the accuracy of the CA method tends to convergence. This method makes it more stable to evaluate the influence of the polarization basis on the estimation of wind and wave parameters.

In the iterative process, we also use three evaluation indicators, RMSE, MAE, and MAPE. A total of 10 000 iterations were performed for each retrieval of the wind and wave

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TABLE III
COR COMPARISON BETWEEN PEAK STATE AND OTHER HIGH COR POLARIZATION BASES

Sea state parameters	The evaluation index	Convergence		Median		Minimum		Variance	
		VV	peak state	VV	peak state	VV	peak state	VV	peak state
SWH	RMSE(m)	0.728	0.655	0.662	0.628	0.514	0.502	0.0032	0.0058
	MAE(m)	0.532	0.474	0.473	0.447	0.378	0.368	0.0199	0.0075
	MAPE(%)	26.70	24.51	24.74	23.56	16.53	17.01	1.36	0.17
WS	RMSE(m/s)	1.894	1.571	1.771	1.512	1.361	1.222	0.1580	0.0479
	MAE(m/s)	1.460	1.180	1.355	0.947	1.046	0.947	0.1120	0.0273
	MAPE(%)	23.30	19.63	21.78	18.97	16.10	14.93	0.28	0.07
SWH(wind)	RMSE(m)	0.769	0.546	0.716	0.523	0.525	0.408	0.0279	0.0097
	MAE(m)	0.530	0.362	0.478	0.343	0.336	0.269	0.0224	0.0056
	MAPE(%)	151.61	103.26	125.17	94.91	69.55	57.24	0.51	0.12
MWP(wind)	RMSE(s)	1.068	0.852	1.012	0.850	0.793	0.687	0.0355	0.0041
	MAE(s)	0.801	0.641	0.746	0.637	0.594	0.505	0.0264	0.0024
	MAPE(%)	23.52	19.38	21.54	19.22	16.92	14.97	0.26	0.02

parameter model. Figs. 17–20 present the results of various wind and wave parameter retrieval models. The black lines correspond to the range on the left y-axis, the colored lines correspond to the range on the right y-axis, and the dashed lines indicate the values of various performance indexes after convergence.

As shown in Fig. 17, after approximately 1000 iterations, RMSE, MAE, and MAPE tend to be convergence. The convergence values of RMSE, MAE, and MAPE obtained using the VV polarization are 0.728 m, 0.53 m, and 26.7%, respectively, and the convergence values of the three indexes obtained using the "peak state" are 0.655 m, 0.474 m, and 24.51%, which increases by 0.073 m, 0.056 m, and 2.19%, respectively. Figs. 18–20 reveal that the polarization basis transformation also improves the ability to retrieve wind-driven wind and wave parameters, and this effect is substantial. Table III shows the statistical value of each index in the iterative process. The results in Table III reveal that, after transforming the polarization basis to "peak state," the performance indexes of the retrieval model have been improved by 10%–20%.

B. Comparison of the Accuracy of Wind and Wave Parameters Retrieval Using Elliptical Polarization Bases With High Correlation Coefficients

As demonstrated in Section III-B, some elliptical polarization bases [e.g., $S_{11}(-80°, -30°)$ and $S_{11}(-70°, -25°)$] have COR values close to the "peak state." According to the comparison of retrieval accuracy between "peak state" and VV polarization, the retrieval capability of "peak state" with high COR is significantly stronger than VV, which indicates that retrieval performance improves with greater correlation. To further verify this issue, the elliptical polarization bases with high COR are selected to retrieve wind and wave information.

In this experiment, the areas of the polarization bases with the COR close to the "peak state" are defined as the "high COR area," such as the adjacent areas of the "peak state." For SWH, WS, SWH (wind), and MWP (wind), four polarization bases, including "peak state," are chosen for comparison and evaluation, as seen in Table III. Figs. 21–24 present the results of wind and wave parameters retrieval models.

As shown in Figs. 21–24, although the polarization basis in the "high COR area" and the "peak state" have different polarization ellipse parameters, there is almost no difference in their retrieval performance. We can also find that the retrieval accuracy of polarization bases with high COR is obviously superior to the commonly used VV polarization, whether it is a mixed wave or wind wave. This demonstrates that the use of elliptical polarization is of great value for wind and wave parameter retrieval, and an appropriate polarization basis can greatly enhance their accuracy.

In addition, there are many "high COR areas" that can be used to improve retrieval performance. As illustrated in

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TABLE IV

STATISTICAL FEATURES OF ITERATIVE INDEXES FOR RETRIEVAL MODELS OF WIND AND WAVE PARAMETERS

Parameters	peak state (ψ,χ)	COR	Polarization basis (ψ,χ)	COR	Polarization basis (ψ,χ)	COR	Polarization basis (ψ,χ)	COR
SWH	$S_{11}(-80°,-25°)$	0.7880	$S_{11}(-70°,-25°)$	0.7879	$S_{11}(-70°,-30°)$	0.7836	$S_{11}(-80°,-30°)$	0.7860
WS	$S_{22}(-80°,-30°)$	0.8095	$S_{22}(-78°,-28°)$	0.8102	$S_{22}(-78°,-30°)$	0.8097	$S_{22}(-80°,-28°)$	0.8089
SWH(wind)	$S_{22}(80°,30°)$	0.8045	$S_{22}(78°,28°)$	0.8048	$S_{22}(78°,30°)$	0.8035	$S_{22}(80°,28°)$	0.8055
MWP(wind)	$S_{22}(80°,30°)$	0.8045	$S_{22}(78°,28°)$	0.8046	$S_{22}(78°,30°)$	0.8039	$S_{22}(80°,28°)$	0.8053

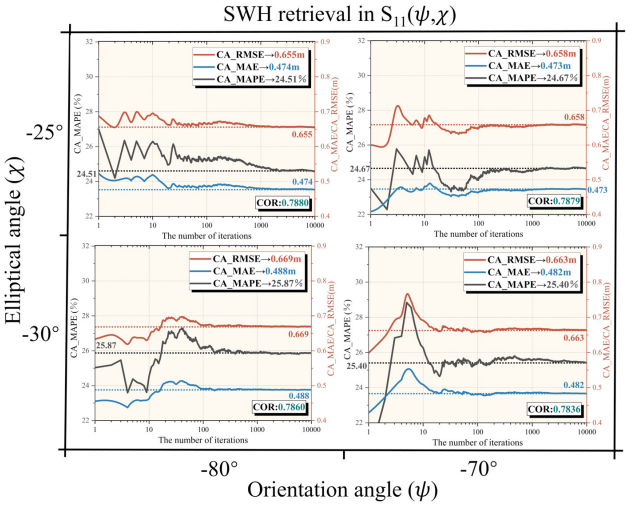

Fig. 21. Iteration diagram of accuracy index of the SWH model retrieval by elliptical polarization bases with strong COR cutoff wavelength (the convergence values of accuracy indexes are shown in the top right, and the COR is marked in the lower right).

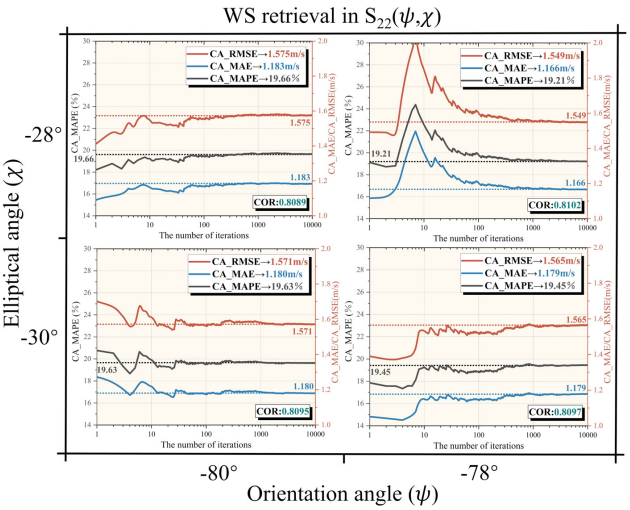

Fig. 22. Iteration diagram of accuracy index of the WS model retrieval by elliptical polarization bases with strong COR cutoff wavelength (the convergence values of accuracy indexes are shown in the top right, and the COR is marked in the lower right).

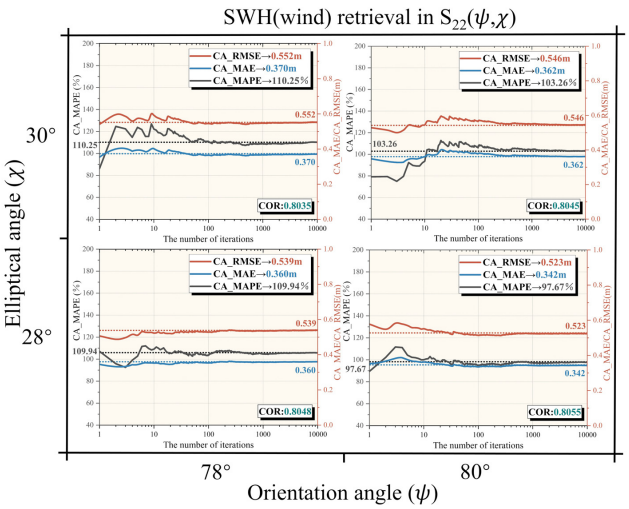

Fig. 23. Iteration diagram of accuracy index of the SWH (wind) model retrieval by elliptical polarization bases with strong COR cutoff wavelength (the convergence values of accuracy indexes are shown in the top right, and the COR is marked in the lower right).

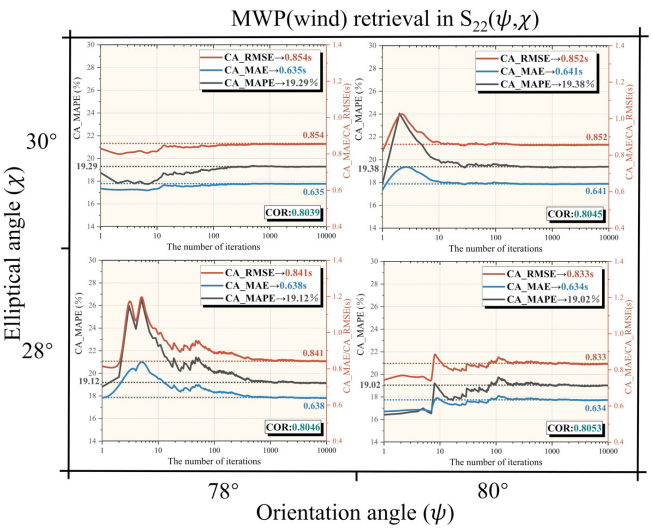

Fig. 24. Iteration diagram of accuracy index of the MWP (wind) model retrieval by elliptical polarization bases with strong COR cutoff wavelength (the convergence values of accuracy indexes are shown in the top right, and the COR is marked in the lower right).

Figs. 25–27, the accuracy comparison of retrieval wind-related parameters under the "peak state" [e.g., $S_{22}(-80°,-30°)$ and $S_{22}(80°,30°)$ and its symmetric polarization basis [e.g., $S_{22}(80°,30°)$, and $S_{22}(-80°,-30°)$] is shown. Although

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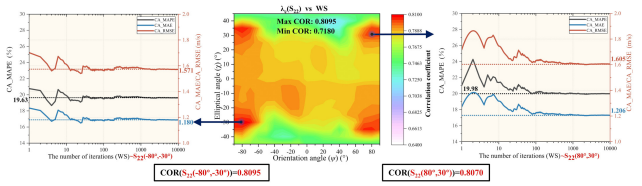

Fig. 25. Iteration diagram of accuracy index of the WS model retrieval by $S_{22}(-80°, -30°)$ and $S_{22}(80°, 30°)$.

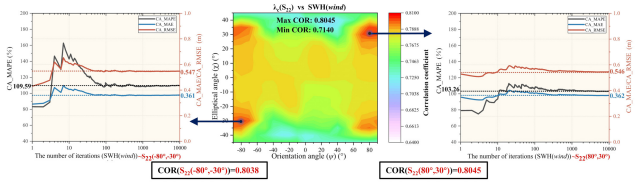

Fig. 26. Iteration diagram of accuracy index of the SWH (wind) model retrieval by $S_{22}(-80°, -30°)$ and $S_{22}(80°, 30°)$.

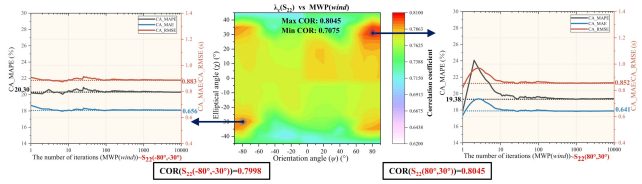

Fig. 27. Iteration diagram of accuracy index of the MWP (wind) model retrieval by $S_{22}(-80°, -30°)$ and $S_{22}(80°, 30°)$.

$S_{22}(80°, 30°)$, and $S_{22}(-80°, -30°)$ have a negative phase difference with the "peak state" [e.g., $S_{22}(-80°, -30°)$ and $S_{22}(80°, 30°)$], both the COR and retrieval performance are significantly improved compared to commonly used H-V polarization base.

VI. Conclusion and Discussion

In the past decade, the potential of polarimetric SAR observations in the study of the imaging physics of upper ocean dynamic radar has been fully demonstrated [45], [46], and the effects of polarization on the cutoff wavelength itself and SWH estimation have been analyzed. However, the impact of polarization state on wind wave information retrieval has not been considered and studied.

In this study, our aim was to explore the influence of different polarization bases on the cutoff wavelength-based retrieval of wind and wave information. GF-3 SAR WM data and ERA-5 forecast data were primarily used to evaluate the correlation between the cutoff wavelength and the wind and wave parameters under the H-V linear, circular, and linear rotated polarization bases. Then, the elliptical polarization base was introduced into the correlation evaluation to analyze the relationship between the cutoff wavelength and marine environment parameters. Finally, the correlation evaluation results were used to compare the retrieval accuracies of wind and wave parameters.

The results show that, only in the special polarization bases states (the H-V linear, circular, and 45° linear rotated polarization), VV polarization has higher COR with wind and wave parameters. The CORs of SWH, WS, SWH (wind), and MWP (wind) can reach approximately 0.77. When the polarization basis is extended to the general elliptical polarization, more suitable polarization basis state [e.g., $S_{11}(-80°, -25°)$, $S_{22}(-80°, -30°)$, and $S_{22}(80°, 30°)$] was found. The correlation between the cutoff wavelengths of these polarization base states and the parameters of the wind and wave can exceed 0.8. It means that the choice of polarization basis is an important factor that cannot be ignored in the retrieval of wind and wave parameters. Compared to VV polarization, the azimuth cutoff wavelength estimated from elliptical polarization bases with maximum COR can reduce the errors of wind and wave parameters retrieval by 10%–20%. In addition, even if the polarization basis having the largest COR is not used, selecting a relatively high COR polarization base can also improve retrieval accuracy.

Our results prove that the H-V linear polarization base is not the optimal solution in the retrieval of wind and wave information. By introducing other polarization bases and reasonably selecting them, the ability of the SAR cutoff wavelength to retrieve wind and wave parameters can be substantially improved.

This study provides guidance for improving the accuracy of retrieval models. In past studies, the optimization of wind and wave parameters retrieval models has stagnated. The establishment of the model is limited to the quad-polarization mode of the original data (HH, HV/VH, and VV), and the potential of the polarization state under other polarization bases has been ignored. Our study has shown that choosing a suitable polarization basis for different wind and wave parameters can effectively improve the accuracy of the retrieval results. This will provide a new reference for future retrieval model optimization.

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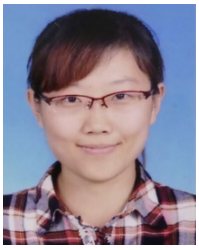

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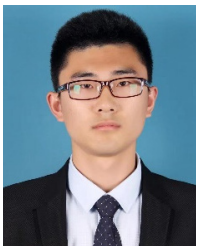

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